Semantic Pattern-based Retrieval of Architectural Floor Plans with Case-based and Graph-based Searching Techniques and their Evaluation and Visualization

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Keywords: Graph Matching, Subgraph Matching, Graph Isomorphism, Architectural Floor Plan, Case-based Reasoning, Pattern Recognition.

Abstract: Until today, for the conceptual design of architectural floor plans, architects widely follow the traditional pen and paper based method to draw the conceptual floor plans, and retrieve the similar floor plans in the printed reference collections. In this paper we present a complete end-to-end system that helps architects to retrieve similar floor plans in early design phases. This work makes a three-fold contribution. Firstly, we have adapted three state of the art techniques to retrieve the similar floor plans: case-based reasoning (CBR), exact graph matching, and inexact graph matching. Secondly, we conducted a test to detect the computational limits of the searching techniques. And finally, we performed a qualitative analysis by running more realistic test cases created by architects while keeping in mind the computational limits. For visualization of results, we have integrated advanced version of our previously implemented web-based user interface. The qualitative analysis showed that the exact graph matching gives in general better results for a majority of test cases, as compared to other two methods. The novelty of our approach is that it combines CBR, exact, and inexact graph matching in one system in the domain of retrieval of architectural floor plans.

1 INTRODUCTION

When starting the design of a building, architects have to develop floor plan concepts facing usually only vague description of the building's requirements. In order to get inspiration to solve the creative problems, working with references is an established method in architecture. Traditionally, this leads to a labor-intensive search and review of magazines and books to find ideas similar to an initial concept. In order to speed up this process, a dedicated search automation would be needed.

For this purpose, we have already presented a distributed case-based retrieval approach *MetisCBR* (Ayzenshtadt et al., 2016a) for search of similar architectural building designs which is prototypically implemented as part of the infrastructure of a basic research project *Metis – Knowledge-based search and query methods for the development of semantic information models (BIM) for use in early design phases*. Metis is an interdisciplinary project, funded by the German Research Foundation (Deutsche Forschungsgemeinschaft, DFG).

MetisCBR can be accessed by a dedicated user interface called *WebUI* (see Figure 1). This web-based GUI allows for creating architectural concepts (floor plans) and sending them to the retrieval engines. The general usability of the WebUI has been shown by the means of a user study (Bayer et al., 2015).

However, so far other types of well-known search methods, like graph and subgraph matching-based retrieval, are not fully tested on this problem, as well as not compared directly with CBR-based approaches.

In this paper we present a complete end-to-end architectural design support system called *Archistant* (http://www.dfki.uni-kl.de/archistant), that implements the previously mentioned WebUI that lets the user draw the required floor plan with rooms and connection between these rooms, adjust the search

⋆Equal contribution to the paper.

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基于语义模式的建筑平面图检索：结合案例推理与图搜索技术及其评估与可视化

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关键词：　图匹配、子图匹配、图同构、建筑平面布局、基于案例的推理、模式识别。

摘要：　迄今为止，在建筑平面方案的概念设计阶段，建筑师仍普遍采用传统的手绘方式绘制概念平面图，并从纸质参考图集中检索相似的平面方案。本文提出了一套完整的端到端系统，旨在帮助建筑师在设计早期阶段检索相似的平面方案。本研究具有三方面的贡献：首先，我们引入并改进了三种先进的检索技术，包括基于案例的推理（CBR）、精确图匹配和非精确图匹配；其次，我们通过实验测定了这些检索技术的计算极限；最后，我们在充分考虑计算资源限制的前提下，利用建筑师提供的更为贴近实际的设计案例开展了定性分析。为便于结果可视化，我们集成了此前开发的基于Web的用户界面的升级版本。定性分析表明，在大多数测试案例中，与另外两种方法相比，精确图匹配通常能够获得更优的检索效果。本研究的创新之处在于，首次将基于案例的推理、精确图匹配与非精确图匹配整合于同一系统中，应用于建筑平面方案的检索领域。

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1 引言

在建筑方案设计之初，建筑师通常只能依据对建筑功能需求的模糊描述来构思平面布局方案。为获取解决创造性问题的灵感，在建筑设计领域，参考既有案例是一种行之有效的方法。传统上，这一过程往往需要耗费大量人力，通过查阅杂志和书籍来寻找与初始概念相近的设计思路。为了加快这一流程，有必要开发专门的搜索自动化工具。

为此，我们已提出了一种分布式基于案例的检索方法*MetisCBR* (Ayzenshtadt等，2016a)，用于搜索相似的建筑方案设计，并已在一项基础研究项目的基础设施中实现了该方法的原型系统 *Metis – Knowledge-based search and query methods for the development of semantic information models (BIM) for use in early design phases* Metis是一项由德国研究联合会（Deutsche Forschungs-gemeinschaft，DFG）资助的跨学科项目。

MetisCBR可通过名为*WebUI* (见图1)的专用用户界面进行访问。该基于Web的图形用户界面允许用户创建建筑方案（平面布置图）并将其发送至检索引擎。通过一项用户研究（Bayer等，2015），已证实该Web界面具有良好的通用可用性。

然而，迄今为止，其他类型的知名检索方法，如基于图匹配和子图匹配的检索方法，尚未在该问题上得到充分验证，也未与基于案例推理的方法进行直接比较。

本文提出了一套完整的端到端架构设计支持系统，名为*Archis- tant* (http://www.dfki.uni-kl.de/archistant)，该系统实现了前述的Web界面，使用户能够绘制包含房间及其相互连接关系的所需平面布局，并对搜索进行调整。

*对该论文的贡献相等。

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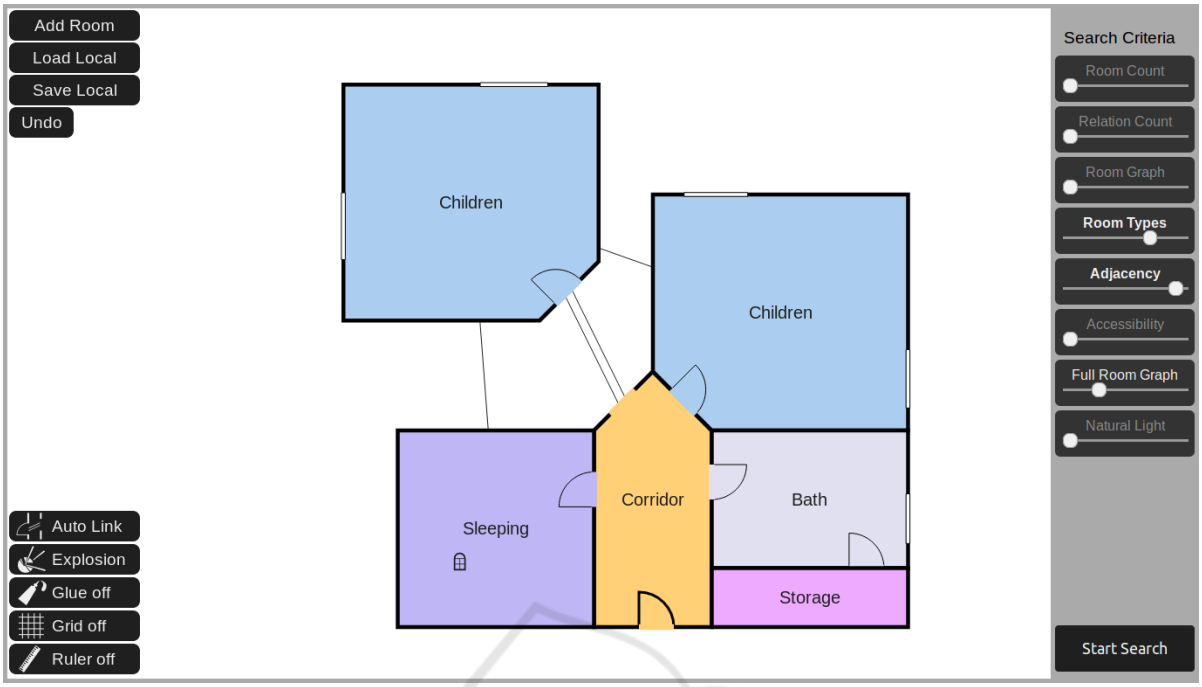


Figure 1: Screenshot of the Archistant WebUI. As a room oriented tool, the individual rooms may be dragged independently from each other, revealing their editable connections to each other. Single lines represent wall connections, double lines represent doors.

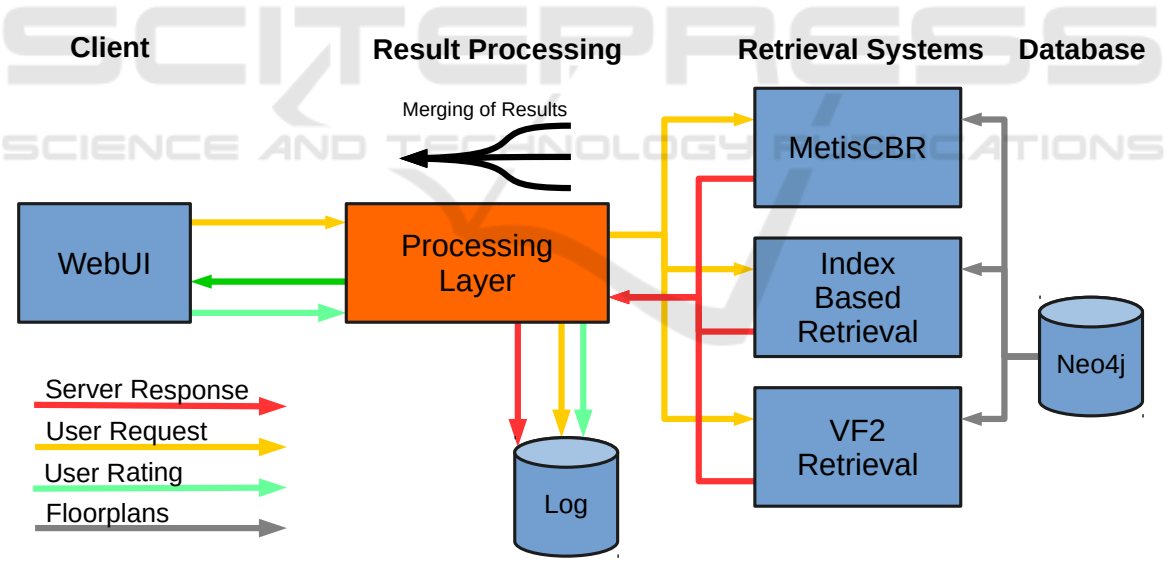


Figure 2: Overview over the system architecture of Archistant (simplified).

criteria, search the similar floor plans, and then visualize the mapping between user's search query (floor plan) and resulted floor plans. For retrieval of similar floor plans, we have implemented three searching techniques, namely case-based retrieval (previously mentioned MetisCBR), exact graph matching, and inexact graph matching. The processing pipeline of Archistant is divided in three steps. First, the user draws the search query with the WebUI. Afterwards the query is forwarded to the search engines, each of them extracts the semantic search patterns *(fingerprints)* (see also Figures 2 and 3) and matches these patterns with floor plans in the database. The result sets of the systems are then unified and sent back to the WebUI for visualization and room mapping.

We also conducted a boundary test of each re-

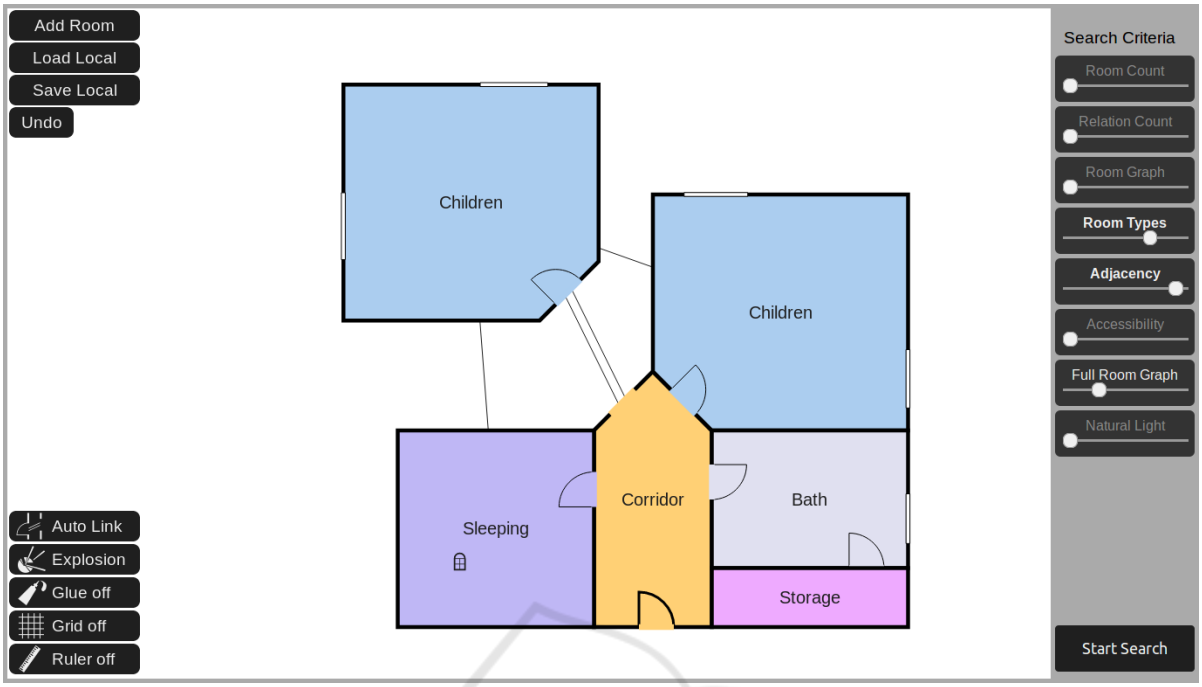


图1：Archistant WebUI 截图。作为一种以房间为单元的工具，各房间可相互独立地进行拖动，从而显示其可编辑的相互连接关系。单线表示墙体连接，双线表示门。

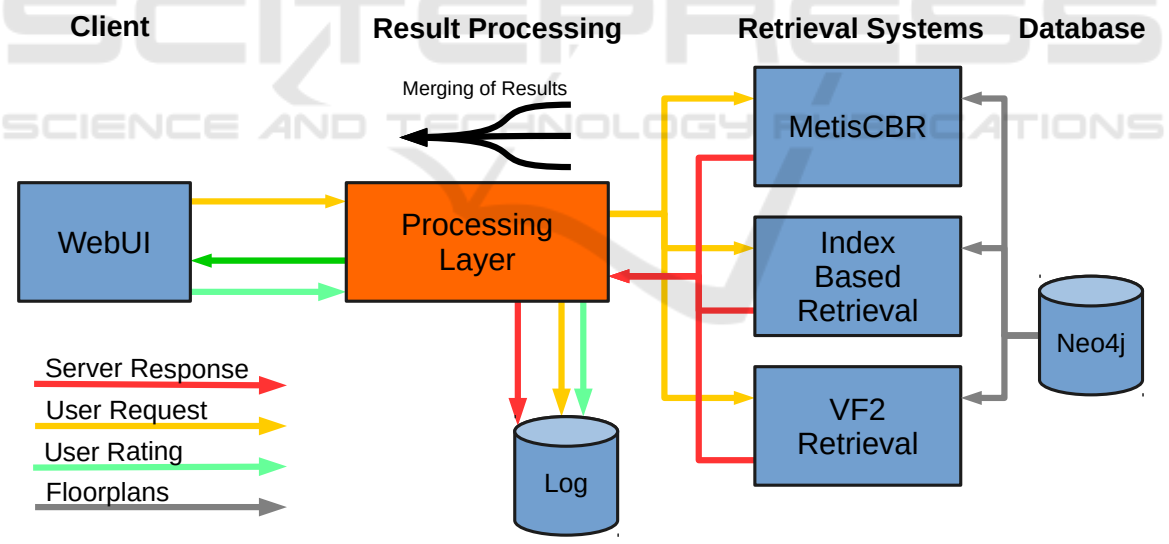


图2：Archistant系统架构概览（简化版）。

标准，搜索相似的平面图，然后可视化用户查询（平面图）与检索结果之间的映射关系。为了实现相似平面图的检索，我们实现了三种检索技术，即基于案例的检索（先前提及的MetisCBR）、精确图匹配和不精确图匹配。Archistant的处理流程分为三个步骤。首先，用户

WebUI生成搜索查询；随后，该查询被转发至各搜索引擎，每个搜索引擎分别提取语义搜索模式*(finger- prints)*（（参见图2和图3），并将其与数据库中的平面布局进行匹配。各系统的检索结果随后被统一，并返回至WebUI以供可视化及房间映射。

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trieval method with respect to each search pattern to analyze their technical limitations. Keeping in mind these technical limitations, we performed a qualitative analysis with some pre-defined search requests to determine which retrieval method is suitable for which user scenario.

2 RELATED WORK

This section provides an overview of work that is related to our paper. Related work can be divided into three categories: CBR, graph and subgraph matching, and sketch-based interfaces.

2.1 Case-based Reasoning

As mentioned in Section 1, MetisCBR uses methods of CBR-based retrieval to find similar floor plans. A comprehensive collections of work related to MetisCBR is described in detail in (Heylighen and Neuckermans, 2001) and (Richter et al., 2007).

2.2 Graph and Subgraph Matching

Graph matching is being widely used these days for the retrieval problems. It might be the case that there is no exact match for the whole graph. To tackle this situation another promising feature is provided that is known as subgraph matching, that is, if two graphs are not completely isomorphic but their part(s) are isomorphic then they can be detected with subgraph matching. An implementation that uses graph matching to retrieve the similar floor plans is described in (Ahmed et al., 2014). This work uses a modified version of (Messmer and Bunke, 1999) for retrieval of similar floor plans by arranging the row-column vectors of the adjacency matrix in the decision tree. A related work in (Wessel et al., 2008) implements graph and subgraph isomorphism to check the similarity between a query graph and building models in the database, considering some constraints, that is, if a query graph corresponds to the constraints, only then it would be checked for similarity.

2.3 Sketch-based Interfaces

In the fields of engineering and architecture, different implementations of the sketch-based interfaces exist that let the user convey the idea by drawing it. A web-based user interface WebUI has been created as part of the previously mentioned research project Metis. It lets the user draw the architectural floor plans electronically, set the different search criteria and, once the results are retrieved, the user can interpret and visualize them with mapping between the query and results (i.e which part of the query relates to which part of the result). For query construction, WebUI uses the AGraphML (Langenhan, 2015) specification.

3 FLOOR PLANS RETRIEVAL TECHNIQUES

To accomplish the task of retrieval of building designs with similar context, we developed a unified retrieval framework (as part of Archistant) with currently three methods implemented. Besides MetisCBR, we have added the integration of two graph-based retrieval techniques: the exact matching method with the VF2 algorithm, and the inexact matching method with the underlying index-based data structure. The main concern we had during the development of the framework was the matter of fact that in real-world scenarios the data is not always consistent, i.e, in some cases the numbers of nodes and/or edges are not equal in the two graphs. This does not allow for determination of the exact isomorphism between these graphs. To deal with such cases, our algorithms should be able to tolerate these errors and instead of finding the only best solution, they should try to find a set of appropriate solutions. This knowledge led us to the implementation of the semantic fingerprint patterns concept in all of the three retrieval methods.

This section is further organized as follows: Section 3.1 defines the *semantic fingerprint*, i.e., the search patterns concept, Section 3.2 defines the graph structure, and then the descriptions of the three matching techniques using these fingerprints are presented in Section 3.3.

3.1 Semantic Fingerprints Concept

The concept of semantic fingerprint of architecture (Langenhan and Petzold, 2010) is based on an index-based hierarchical structure of building data. The hierarchical nature of semantic fingerprints allows for abstract representation of building topology and relational connections between its parts. A fingerprint pattern is a predefined prototype of such a structure and is represented as a graph (see Figure 3) with options to have node labels (names of rooms) and edge lables (names of edges/connections between rooms).

By using the idea of semantic fingerprint, our retrieval methods are able to either decompose (graph and subgraph matching based techniques) the user query into search patterns (fingerprints) or to derive

针对每种检索模式，我们采用相应的检索方法对其技术局限性进行了分析。在充分考虑这些技术局限性的基础上，我们基于若干预定义的检索请求开展了定性分析，以确定哪种检索方法适用于何种用户场景。

2 相关工作

本节概述了与本文相关的研究工作。相关研究可分为三大类：基于案例的推理、图与子图匹配，以及基于草图的交互界面。

2.1 基于案例的推理

如第1节所述，MetisCBR采用基于案例推理的检索方法来寻找相似的平面布局。与MetisCBR相关的研究成果已在（Heylighen和Neucker-mans，2001）以及（Richter等，2007）中得到了详尽的阐述。

2.2 图与子图匹配

图匹配技术如今已被广泛应用于检索问题。在实际应用中，往往会出现整幅图不存在精确匹配的情况。为应对这一情形，一种颇具前景的补充机制——子图匹配应运而生：当两幅图并非完全同构，但其部分子结构彼此同构时，便可借助子图匹配加以识别。文献（Ahmed等，2014）描述了一种基于图匹配实现相似平面图检索的方案；该方法通过将邻接矩阵的行向量与列向量组织于决策树中，采用（Messmer和Bunke，1999）的改进版本来完成相似平面图的检索。另一项相关工作（Wessel等，2008）则利用图同构与子图同构技术，在满足特定约束条件的前提下，对查询图与数据库中的建筑模型进行相似性比对：只有当查询图符合这些约束条件时，才会进一步开展相似性检测。

2.3 基于草图的交互界面

在工程与建筑领域，已存在多种基于草图的交互界面实现，使用户能够通过绘制来表达其设计构想。作为前述Metis研究项目的一部分，开发了一种基于Web的用户界面（WebUI），该界面允许用户以电子方式绘制建筑平面图，并设置各项检索条件，随后……

在检索出结果后，用户可以通过查询与结果之间的映射关系来对其进行解释和可视化（即明确查询的哪些部分与结果的哪些部分相关）。在查询构建方面，WebUI采用AGraphML（Langenhan，2015）规范。

3 楼层平面图检索技术

为实现具有相似语境的建筑设计检索任务，我们开发了一个统一的检索框架（作为Archistant的一部分），目前已实现了三种方法。除MetisCBR外，我们还集成了两种基于图的检索技术：一种是采用VF2算法的精确匹配方法，另一种是基于底层索引数据结构的近似匹配方法。在框架开发过程中，我们主要关注的一个问题是，在实际应用场景中，数据往往并不一致，即在某些情况下，两个图的节点数和/或边数并不相等。这使得无法确定这两者之间是否存在精确的同构关系。为此，我们的算法必须能够容忍这些误差，并且不应仅寻求唯一的最优解，而应尝试找出一组合适的解。这一认识促使我们在所有三种检索方法中都引入了语义指纹模式的概念。

本节的组织结构如下：第3.1节定义了*semantic fingerprint*，即搜索模式的概念；第3.2节定义了图结构；随后，第3.3节将介绍基于这些指纹的三种匹配技术。

3.1 语义指纹概念

建筑语义指纹的概念（Langenhan 和 Petzold，2010）基于一种以索引为基础的建筑数据层次化结构。语义指纹的层次性特征使其能够对建筑拓扑及其各组成部分之间的关联关系进行抽象表达。指纹模式是此类结构的预定义原型，以图的形式表示（见图3），并可为节点和边分别设置标签：节点标签用于标识房间名称，边标签用于标识房间之间的连接关系。

通过采用语义指纹的思想，我们的检索方法能够将用户查询分解为搜索模式（指纹），或者从中推导出

the fingerprints from the query (MetisCBR), and then match them with other floor plans in our database.

AGraphML is the XML-based representation for floor plans. The specification (Langenhan, 2015) of AGraphML defines attributes for this representation. It can represent a complete architecture for a floor plan, including the nodes and edges with their attributes.

Figure 3 shows a list of 7 fingerprints currently implemented in each of our retrieval engines.

3.2 Query Structure

Search queries and floor plans in our database are represented as graph $G = (V, E)$. Here, V represents the vertices (nodes) in the graphs, which are rooms in our case, e.g., Kitchen or Sleeping. E represents the edges (the connections between nodes), e.g., Kitchen and Sleeping are connected with an edge called Door.

3.3 Matching Techniques

This section presents the three matching and retrieval techniques implemented in Archistant: MetisCBR, exact graph matching (referred as *VF2*), and inexact graph matching (referred as *index-based*).

3.3.1 MetisCBR

MetisCBR is based on a decentralized structure where agents and agent groups (retrieval containers) work on different, predefined and assigned, tasks to support an architect during the early design phases. The actual retrieval process is coordinated by a special agent group *(MetisCBR Coordinator)* described in (Ayzenshtadt et al., 2016b). Being a case-based retrieval system, MetisCBR's theoretical emphasis is built on the basic CBR paradigm that similar problems have similar solutions. The system analyses the given problem (user's search request) and tries to find solutions (search results) that can be the most helpful to solve this problem, that is, adapt the currently created design according to findings in the search result. MetisCBR is related to other systems and approaches that were developed for the purpose of assisting an architect, some examples are FABEL (Voss, 1997), CBArch (Cavieres et al., 2011), or CaseBook (Inanc, 2000). To define a case (architectural design graph) within the system, a specific distributed model (see Figure 4) is applied to each graph-based floor plan. The floor plans are then imported into the case base of the system, where they are separated into three main concepts: floor plan meta data, rooms and edges. Each of the concepts consists of a number of specific attributes (see also (Ayzenshtadt et al., 2015)). During the search, the weighting of single attributes and an amalgamation of them is applied to find the most similar concepts for the given search request (or its parts).

The amalgamations are divided into fingerprint-related and generic. The fingerprint-related amalgamations are applied only if one or more semantic fingerprint patterns were added to the query. Each of the fingerprint amalgamations considers only attributes that were defined for this particular pattern. This process is related to the footprint sets-based retrieval described in (Smyth and McKenna, 1999). In contrast to the fingerprint-related, the generic amalgamation functions use all of the attributes, and are applied if no fingerprint definitions were found. Both of the amalgamation types can be executed within different retrieval strategies. Currently, two kinds of strategies are implemented in the system: a multistep results preselection-based strategy for complex fingerprint patterns as well as for the deep search without fingerprints (described in (Ayzenshtadt et al., 2015)), and a single-step strategy for general retrieval with or without fingerprints . A post-retrieval weighting process (available only for search requests with multiple fingerprints) can be applied to boost results for fingerprints that were considered more important by the user. The results of the retrieval are returned sorted in descending order by the computed confidence score.

3.3.2 Exact Graph Matching Method (VF2)

In exact graph matching, one-to-one mapping is known as isomorphism. When two graphs contain the same number of nodes, and they are connected in the same way, then we can call them isomorphic. Isomorphism can be found with exact graph matching (Bengoetxea, 2002). For example, (Ullmann, 1976), (Schmidt and Druffel, 1976), and (McKay et al., 1981) are one-to-one exact graph-based matching approaches. Our system (Archistant) uses implementation of the VF2 algorithm, proposed in (Cordella et al., 2004) (and implemented in the NetworkX library), because it significantly reduces the memory requirement and obtains the best performance for graphs of small size and for quite sparse graphs (Foggia et al., 2001).

In our system, VF2-based method uses AGraphML files to generate the graphs. Firstly, AGraphML files from Neo4j database are generated, as an offline step, with a tool named "Neo4j Shell Tools". One AGraphML file will be generated against each floor plan. Now our system can match the search request with other floor plans of Neo4j database.

The step by step details of how VF2 system

首先提取查询样本的特征指纹（MetisCBR），然后将其与数据库中的其他平面布局进行匹配。

AGraphML 是用于楼层平面图的基于 XML 的表示形式。AGraphML 的规范（Langenhan，2015）为此表示形式定义了相关属性。它能够完整地描述一个楼层平面图的架构，包括节点及其属性和边及其属性。

图3展示了目前我们各检索引擎中已实现的7种指纹特征列表。

3.2 查询结构

我们数据库中的搜索查询和楼层平面图均以图结构 $G = (V, E)$ 表示。其中，V 表示图中的顶点（节点），在本场景中即为房间，例如厨房或卧室；E 表示边（节点之间的连接关系），例如厨房与卧室之间通过一条名为“门”的边相连。

3.3 匹配技术

本节介绍了Archistant中实现的三种匹配与检索技术：MetisCBR、精确图匹配（记为*VF2*）以及非精确图匹配（记为*index-based*）。

3.3.1 MetisCBR

MetisCBR采用去中心化架构，其中各智能体及智能体组（检索容器）分别承担预先定义并分配的任务，以支持建筑师在设计的早期阶段开展工作。实际的检索过程由一个专门的智能体组*(MetisCBR Coordinator)*协调完成，该组的相关描述见（Ayzenstadt等，2016b）。作为一种基于案例的检索系统，MetisCBR的理论基础建立在基本的CBR范式之上，即相似的问题往往具有相似的解决方案。该系统首先对给定问题（用户的检索请求）进行分析，并尝试寻找能够最有效地解决该问题的方案（检索结果），亦即根据检索结果中的发现对当前设计进行相应调整。MetisCBR与其他旨在辅助建筑师的设计系统或方法存在关联，例如FABEL（Voss，1997）、CBArch（Caviares等，2011）以及CaseBook（Inanc，2000）。为在系统中定义一个案例（建筑方案图），针对每个基于图模型的平面布置图均采用一种特定的分布式表示模型（参见图4）。随后，这些平面布置图被导入系统的案例库，并按三个主要概念进行划分：平面布置元数据、房间与边。每个概念均由若干具体属性（另见 Ayzenshtadt 等人，2015）。在检索过程中，通过对单个属性及其组合进行加权，以确定与给定检索请求（或其组成部分）最为相似的概念。

合并操作可分为与指纹相关的合并和通用合并两类。仅当查询中添加了一个或多个语义指纹模式时，才会应用与指纹相关的合并；每种与指纹相关的合并仅考虑为该特定模式所定义的属性。这一过程与文献（Smyth 和 McKenna，1999）中所述的基于足迹集的检索方法相关。与之相对，通用合并则使用所有属性，并在未发现任何指纹定义时予以应用。两种类型的合并均可在不同的检索策略下执行。目前，系统中已实现两类策略：一种是针对复杂指纹模式以及无指纹深度搜索的多步结果预选策略（参见 Ayzenshtadt 等人，2015），另一种是适用于带或不带指纹的一般检索的单步策略。对于包含多个指纹的搜索请求，还可进一步采用后置加权处理，以提升用户认为更为重要的指纹所对应的结果权重。检索结果按计算得到的置信度得分由高至低排序后返回。

3.3.2 精确图匹配方法（VF2）

在精确图匹配中，一对一映射被称为同构。当两个图的节点数相同且连接方式一致时，我们称它们为同构图。同构关系可通过精确图匹配来识别（Bengoetxea, 2002）。例如，Ullmann（1976）、Schmidt和Druffel（1976）以及McKay等人（1981）均属于基于一对一精确图匹配的方法。我们的系统（Archistant）采用了Cordella等人（2004）提出的VF2算法的实现方案（该算法已在NetworkX库中得到具体实现），因为该算法能够显著降低内存需求，并在小规模图以及较为稀疏的图上表现出最佳性能（Fogia等人，2001）。

在我们的系统中，基于VF2的算法采用AGraphML文件来生成图结构。首先，通过名为“Neo4j Shell Tools”的工具，在离线阶段从Neo4j数据库中导出AGraphML文件；每个楼层平面图对应生成一个AGraphML文件。由此，本系统即可将查询请求与Neo4j数据库中的其他楼层平面图进行匹配。

VF2系统的逐步详细说明

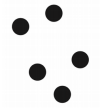
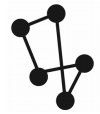
Fingerprint	Name	Description	Specifics
FP1 	Room Count	Number of rooms	No connections between rooms and no labels specified
FP2 	Relation Count	Number of edges	No room information specified
FP3 	Room Graph	Anonymous representation of rooms and edges	No labels specified for rooms and edges
FP4 	Room Types	Labels of rooms	No connections between rooms only room labels are specified
FP5 	Adjacency	Emphasis on room semantics	Rooms information is complete no edge labels specified
FP6 	Accessibility	Emphasis on edge semantics	Edge information is complete no room labels specified
FP7 	Full Graph	Complete graph	All information about rooms and edges available

Figure 3: Fingerprints (i.e., search patterns) allow for abstract representation of building topology and relational connections between its parts. Fingerprints currently implemented in all examined retrieval methods are shown here. Nodes represent the rooms, edges represent the room connections.

matches (see Figure 5) search request with floor plans is described as follows: in the first step, the system receives the search request, then it checks for its validity, the system proceeds if the request is valid. Then, from the search request, AGraphML is extracted to generate a graph, referred as *query-graph*, and subsequently its fingerprints are generated. After having fingerprints for query-graph, VF2 system then one by one takes each of the offline generated AGraphML files, generates its graph, referred as *db-graph*, and its fingerprints, and then matches the corresponding fingerprints, i.e., FP1 of the query-graph will be matched against FP1 of db-graph of each AGraphML, FP2 of query-graph with FP2 of db-graph of each AGraphML and so on. Once the system is done with matching fingerprints, it delivers the resulting floor plans with confidence score in descending order, the confidence score shows how closely a floor plan matches with the user's search request according to each of the fingerprints in Figure 3.

3.3.3 Inexact Graph Matching Method (Index-based)

There are different types of inexact, index-based graph searching methods: GraphGrep (Giugno and Shasha, 2002), Lucene index (Sharanya Jayaraman, 2013), FG-Index (Cheng et al., 2007), cIndex (Chen et al., 2007). In this paper, we are using the Lucene index-based method as it comes with the Neo4j framework by default.

The index-based method operates as follows (see Figure 6): A search request is decomposed into the different fingerprints. The fingerprints are represented as *Cypher* (Neo4j query language) queries and the ones whose weight is different from 0 are transmitted to the Neo4j server. The Neo4j server answers the requests with a set of floor plans for each finger-

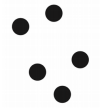
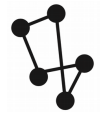
Fingerprint	Name	Description	Specifics
FP1 	Room Count	Number of rooms	No connections between rooms and no labels specified
FP2 	Relation Count	Number of edges	No room information specified
FP3 	Room Graph	Anonymous representation of rooms and edges	No labels specified for rooms and edges
FP4 	Room Types	Labels of rooms	No connections between rooms only room labels are specified
FP5 	Adjacency	Emphasis on room semantics	Rooms information is complete no edge labels specified
FP6 	Accessibility	Emphasis on edge semantics	Edge information is complete no room labels specified
FP7 	Full Graph	Complete graph	All information about rooms and edges available

图3：指纹（即检索模式）能够对建筑拓扑结构及其各组成部分之间的关联关系进行抽象化表征。此处展示了所有已考察的检索方法中当前所采用的指纹表示。节点代表房间，边代表房间之间的连接。

将搜索请求与楼层平面图进行匹配（见图5）的过程描述如下：首先，系统接收搜索请求，并对其有效性进行检查；若请求有效，则继续执行。随后，从搜索请求中提取AGraphML数据以构建一张图，记为*query-graph*，并进一步生成其指纹特征。在获得查询图的指纹特征后，VF2系统依次调用预先离线生成的各AGraphML文件，分别构建相应的图，记为*db-graph*，并生成其指纹特征，然后对相应指纹进行匹配，即查询图的FP1与每个AGraphML数据库图的FP1进行比对，查询图的FP2与每个AGraphML数据库图的FP2进行比对，依此类推。当指纹匹配完成后，系统按置信度得分降序输出匹配结果，其中置信度得分反映了某一楼层平面图与用户搜索请求的匹配程度。

对图3中的每一种指纹进行分析。

3.3.3 不精确图匹配方法（基于索引）

存在多种基于索引的不精确图搜索方法，包括：GraphGrep（Giugno和Shasha，2002）、Lucene索引（Sharanya Jayaraman，2013）、FG-Index（Cheng等，2007）以及cIndex（Chen等，2007）。在本文中，我们采用的是基于Lucene索引的方法，因为该方法默认随Neo4j框架一同提供。

基于索引的方法按如下方式运行（见图6）：搜索请求被分解为不同的指纹。这些指纹以*Cypher*（Neo4j查询语言）表示，且权重不为零的指纹会被发送至Neo4j服务器。Neo4j服务器针对每个指纹返回一组楼层平面图。

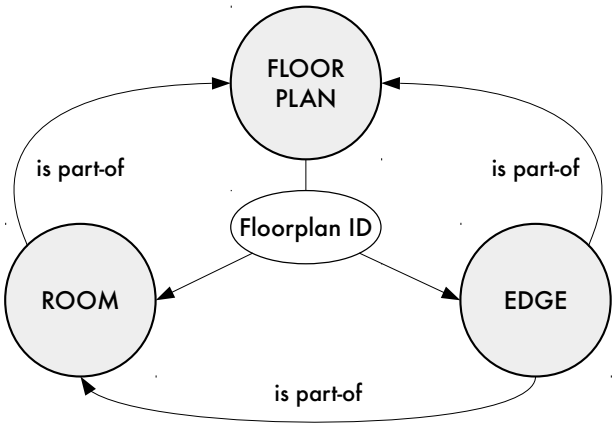


Figure 4: The general description of the underlying distributed model for case-based retrieval in MetisCBR.

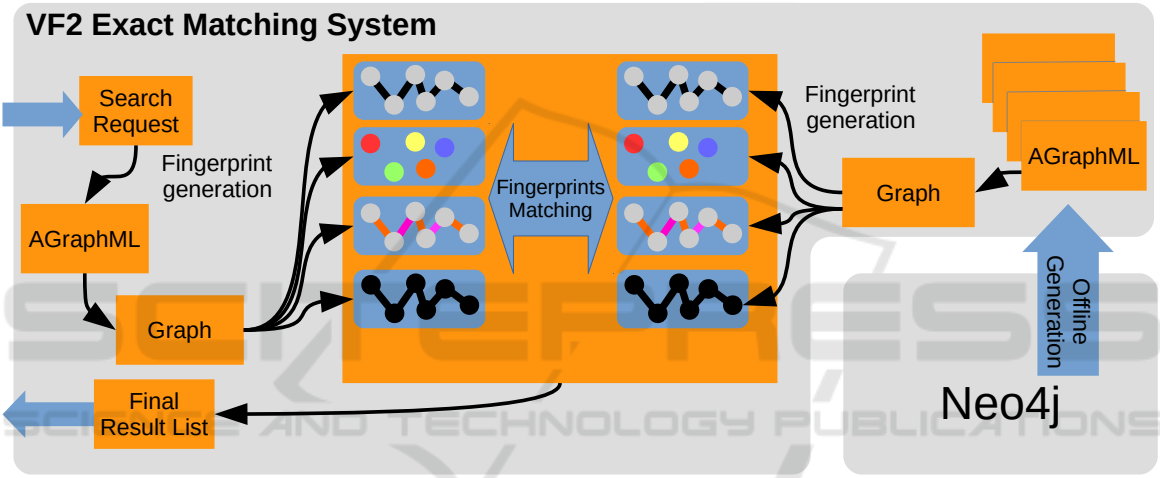


Figure 5: Above is the flow diagram of the VF2 exact matching system. It shows that from the search request first the AGraphML, then the graph and finally the fingerprints are generated. In the next step, the search request's fingerprints are matched with the corresponding fingerprints of the offline generated AGraphML files. Finally, the results are transferred to the requester.

print. These sets are unified and the floor plan results are ordered according to the similarity score which is calculated for every item in the final result set. This similarity score is the sum of the user-defined weights of the fingerprints for which the query matches the database entry.

A fingerprint is considered to match if the fingerprints graph is the subgraph of the database entry. The fingerprints are processed independently for simplicity reasons, hence one room in the query may be mapped to different rooms within the same floor plan in the database. Figure 7 illustrates an example of the fingerprints processing within the index-based method. The query consists of three rooms labeled as Living, Kitchen and Sleeping. The Living room is connected with Kitchen via an edge connection labeled as Passage, the Kitchen is connected with Sleeping via an edge connection labeled as Wall, and Sleeping room is connected with Living via an edge connection labeled as Door. The right side of the diagram shows exemplary matching and non matching fingerprints between search query and floor plan in the database.

4 EVALUATIONS OF OUR SYSTEM

4.1 Technical Limitations (Boundary Test)

Because of the computational demands of matching and other problems, all retrieval methods have their

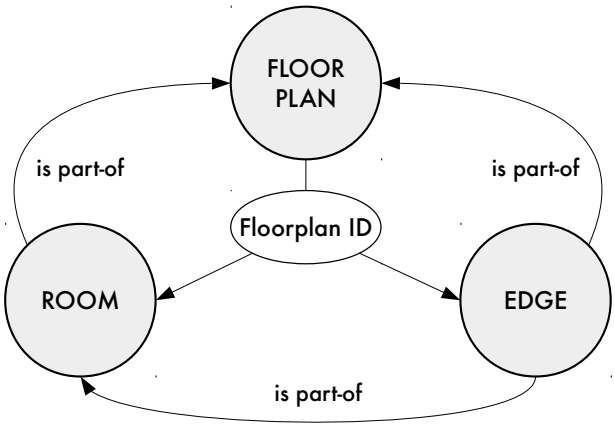


图4：MetisCBR中基于案例检索的底层分布式模型的总体描述。

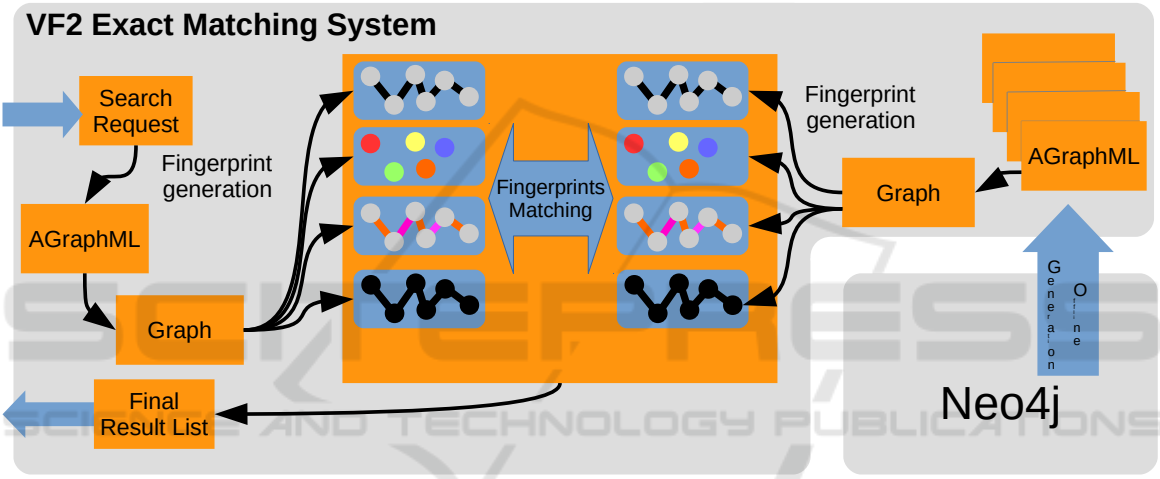


图5：上方为VF2精确匹配系统的流程图。该图显示，从搜索请求开始，首先生成AGraphML，随后生成图结构，最后生成指纹。在下一步中，将搜索请求的指纹与离线生成的AGraphML文件中的相应指纹进行匹配。最后，将结果返回给请求方。

打印。这些集合是统一的，且平面布置图结果按照相似度得分进行排序，该得分是针对最终结果集中每个条目计算得出的。所述相似度得分即为查询与数据库条目匹配的指纹所对应的用户定义权重之和。

当指纹图是数据库条目的子图时，即认为指纹匹配。为简化起见，指纹处理是独立进行的，因此查询中的某一房间可能会映射到数据库中同一楼层平面图内的不同房间。图7展示了基于索引的方法中指纹处理的一个示例。查询由三个房间组成，分别标记为客厅、厨房和卧室。客厅通过一条标注为“通道”的边连接到厨房，厨房则与……

通过标为“Wall”的边连接进入“Sleeping”状态，而“Sleeping room”则通过标为“Door”的边与“Living”相连。图的右侧展示了搜索查询与数据库中楼层平面之间的示例性匹配与不匹配指纹。

4 我们系统的评估

4.1 技术局限性（边界测试）

由于匹配及其他问题带来的计算开销，所有检索方法都存在其

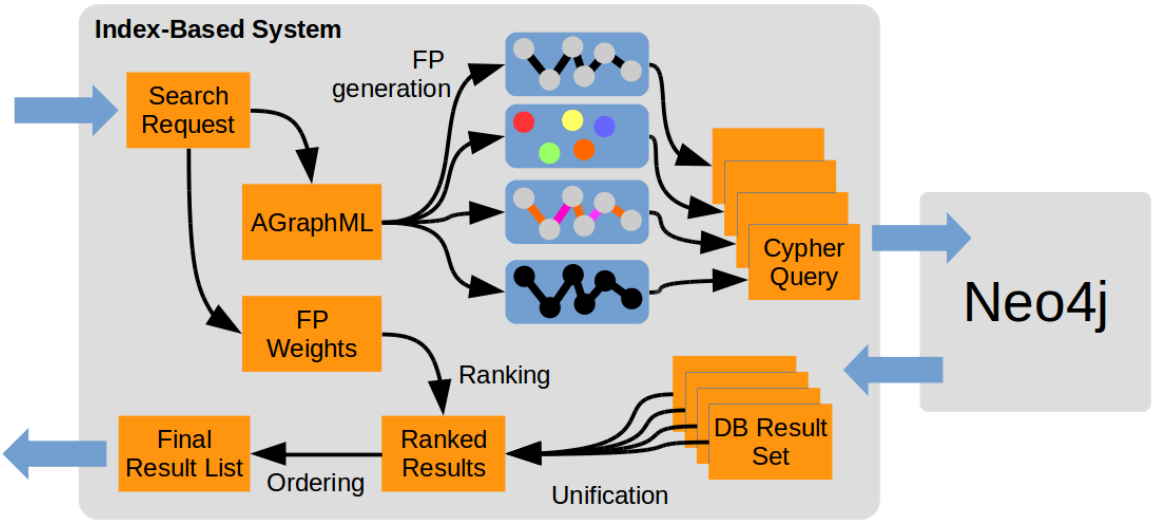


Figure 6: The index-based method flow diagram. A search request is decomposed into AGraphML and fingerprint weights. From the AGraphML the fingerprints are created and the fingerprints are translated into Cypher queries. The Cypher queries are send to the database, each Cypher-translated fingerprint is answered with a set of floor plans. The result sets are unified. Each floor plan is scored by the sum of the weights of the fingerprints corresponding to the the result sets that contain the floor plan. Finally, the result set is sorted by the score and transferred to the requester.

own limitations regarding the complexity of search requests they can handle, especially with respect to each of the fingerprints shown in Figure 3. Given enough time and storage, all search queries should be answerable in theory, but since computation time must be limited due to usability reasons and storage is limited due to economical reasons, the question arises how complex the search requests may be so that they can be reasonably processed by a given system. In order to test our retrieval methods, we employed the following method: for each type of fingerprint we generated test scenarios of variable complexity which we instantiated over an interval between 1 and 25 units of complexity, where 25 is selected empirically as a maximum limit. The metrics for these units differ for the different fingerprints, but are either equal to the number of rooms in a test scenario or the number of edges in a test scenario. The test scenarios are designed so that the systems are advised to only take the fingerprint into account the scenario was designed for. The most complex test scenario that could be handled by the system is considered to be the system's boundary for that fingerprint. The results are shown in Figure 8 that contains the boundaries of the fingerprints we tested on our systems. Both the VF2 exact matching and MetisCBR executed all test scenarios flawlessly, only the index-based system has limitations over the maximum size of fingerprints FP3, FP5, FP6 and FP7. These limitations came from internal timeout errors of the underlying Neo4j database, most likely due to a congestion of the database system. Knowing the boundaries of our different methods for a given hardware and time limitation, we defined test scenarios for a subsequent, qualitative analysis (see Section 4.2) in a more safe manner.

4.2 Qualitative Analysis

The evaluation of quality and helpfulness of the retrieval results of the above described retrieval methods was the main task of the subsequent special study, where the mixed team of architects and computer scientists created queries and rated collaboratively a number of graphical and graph-based result representations of the search result sets. The confidence scores (range $R = 0.0..1.0$), computed by the retrieval methods during the retrieval, were taken into account as well.

As search queries, 10 AGraphML requests with floor plan constructions of different complexity, were used. Each of the presented retrieval methods was queried with each of those AGraphML-based floor plans and delivered accumulated result sets, sorted in descending order by confidence score of the single result. All of the fingerprint patterns were applied to the given queries. For this qualitative analysis we asked different participants to observe the results of each of the three retrieval methods, corresponding to each of the 10 queries. Each participant rated the three methods on the scale of first, second and third or equal. In Table 1 the results of the subjective qualitative analysis from the participants of the study are shown. This

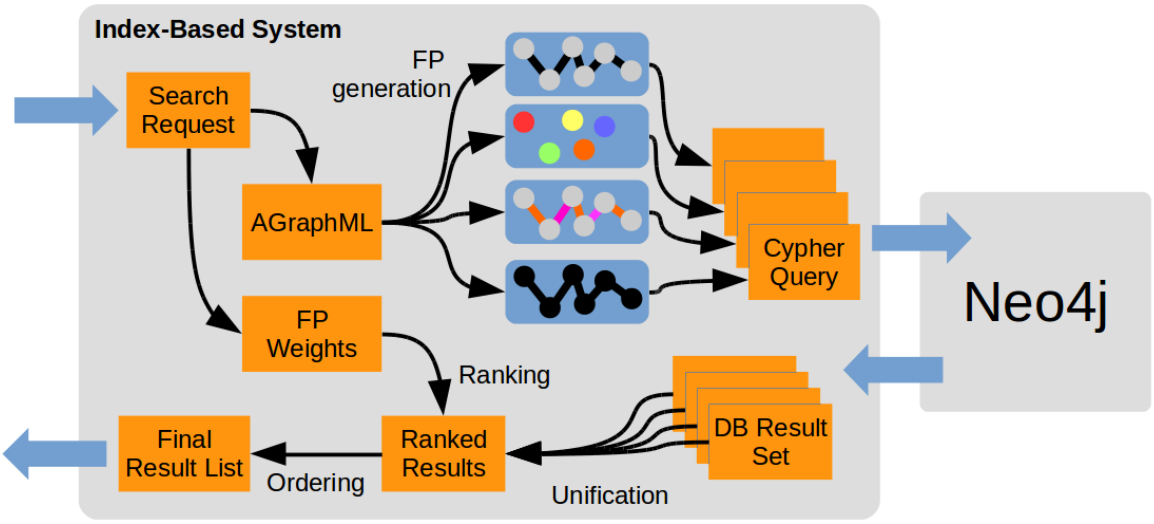


图6：基于索引的方法流程图。搜索请求被分解为AGraphML和指纹权重。从AGraphML中生成指纹，并将这些指纹转换为Cypher查询。Cypher查询被发送至数据库，每个经Cypher转换的指纹均会返回一组楼层平面图。各结果集被合并统一。每个楼层平面图的得分由包含该楼层平面图的结果集中相应指纹的权重之和确定。最后，结果集按得分排序后返回给请求者。

各系统在所能处理的查询复杂度方面均存在自身局限，尤其是在针对图3所示的各种指纹类型时。理论上，只要时间与存储资源充足，所有搜索查询皆可得到解答；然而，出于可用性考虑必须对计算时间加以限制，同时受经济因素制约存储空间亦有限，因此亟需明确：给定系统能够合理处理的查询复杂度上限究竟为多少？为检验我们的检索方法，我们采用了如下策略：针对每种指纹类型，我们构建了不同复杂度的测试场景，并将其复杂度参数在1至25个单位之间进行递增设置，其中25被经验性地选定为上限值。各单位所对应的度量指标因指纹类型而异，或以测试场景中的房间数表示，或以测试场景中的边数表示。这些测试场景的设计原则是：系统仅应依据为其量身定制的特定指纹开展匹配。对于某一指纹而言，系统所能处理的最复杂测试场景即被视为其相应边界。结果如图8所示，其中展示了我们在各系统上测试的各类指纹的处理边界。VF2精确匹配算法与MetisCBR均能无误地执行所有测试场景，唯有基于索引的系统在FP3、FP5、FP6及FP7等指纹的最大规模处理上存在局限。这些局限源于底层Neo4j数据库的内部超时错误，很可能系由数据库系统的负载过饱和所致。鉴于此，为了在给定的硬件条件和时间限制下明确各方法的适用范围，我们以更为稳妥的方式定义了用于后续定性分析的测试场景（详见第4.2节）。

4.2 定性分析

对上述检索方法所生成的检索结果的质量与有用性进行评估，是后续专项研究的主要任务。在该研究中，由建筑师与计算机科学家组成的跨学科团队共同构建查询，并协作对一系列图形化及基于图的检索结果集表示形式进行评分。此外，还考虑了各检索方法在检索过程中计算得到的置信度得分（取值范围为$R = 0.0..1.0$）。

作为搜索查询，我们使用了10个具有不同复杂度的楼层平面结构的AGraphML请求。针对每一个所呈现的检索方法，均以这10个基于AGraphML的楼层平面作为查询输入，并输出按单个结果置信度得分降序排列的累积结果集。所有指纹模式均被应用于这些查询。在本次定性分析中，我们邀请不同的参与者观察三种检索方法分别对应于10个查询的结果，并按照第一、第二、第三或并列的等级对这三种方法进行评分。表1展示了研究参与者的主观定性分析结果。此

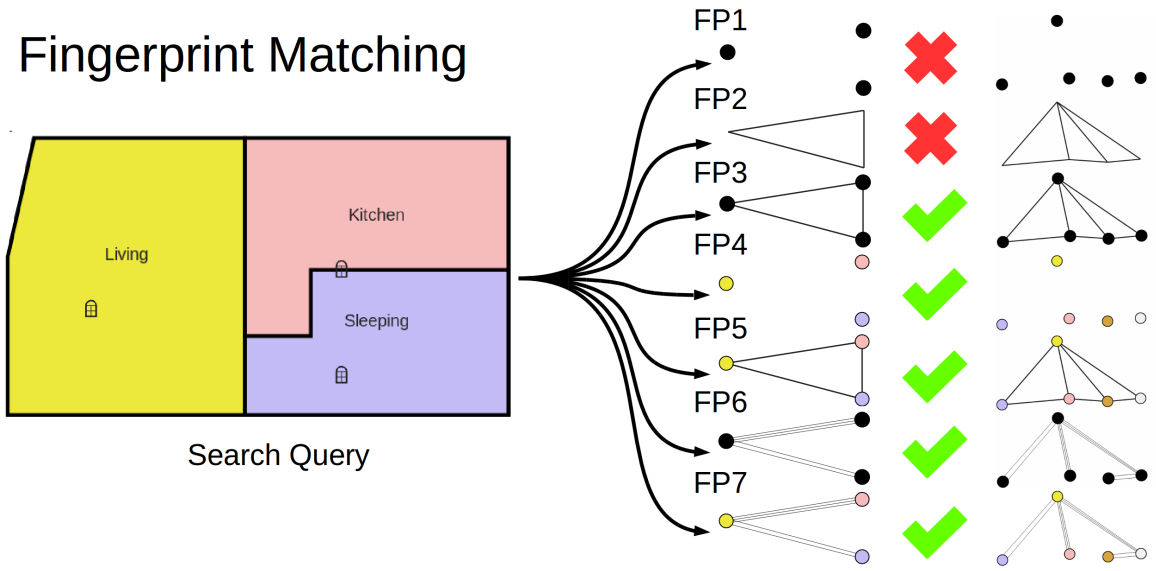


Figure 7: The above diagram shows how the fingerprints are integrated in the retrieval process. The right side of the diagram indicates which fingerprints of the search query are matched with which fingerprints of the floor plan in the data base. Cross indicates no match while green sign indicates a match. The diagram illustrates this for index based method, which also considers the inexact matching.

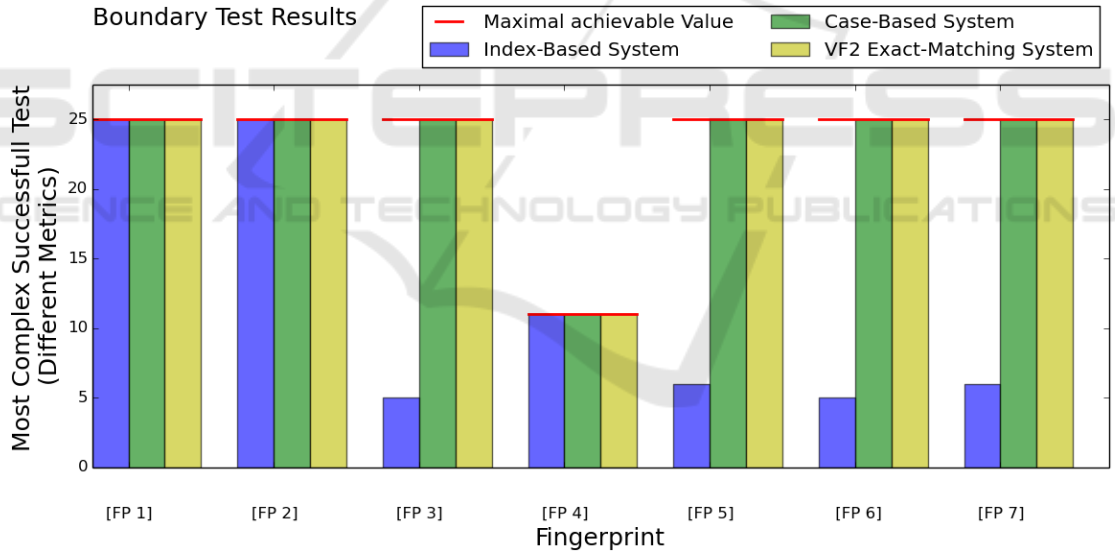


Figure 8: Boundary Test Results. Depicted are the results of the boundary test for each search technique and fingerprint. More precisely, for every combination of search technique and fingerprint the diagram shows the complexity rating of the most complex test scenario the system could handle successfully given a machine with certain computational and memory limits. The metric of complexity differs for the test scenarios of different fingerprints.

table accumulates the overall rankings and presents the summarized results for each query. The ranking results are divided into two categories, where the first category represents queries for which the corresponding retrieval method was determined as the clear winner, that is, was considered best by all participants. The second category is similar to the first one, but shows which method won the majority of rankings for the particular query. In case of equality of votes, the corresponding query is placed in the third column. In the last column the percentage of queries dominated by the particular method is given.

Table 1 shows that each of the three retrieval methods is good for some query cases. In Figure 9, we show the two best results of three selected queries for each search method, where for each of the three se-

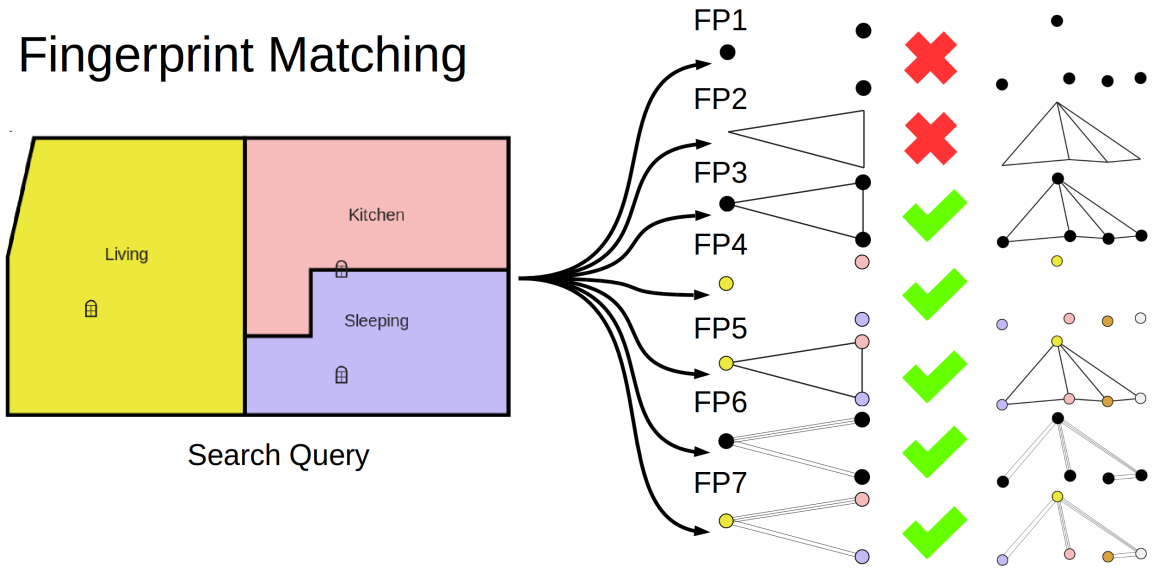


图7：上图展示了指纹在检索过程中的集成方式。图的右侧标明了查询指纹与数据库中楼层平面图指纹之间的匹配情况，其中“×”表示不匹配，绿色标记表示匹配。该图以基于索引的方法为例进行了说明，该方法同时考虑了不精确匹配。

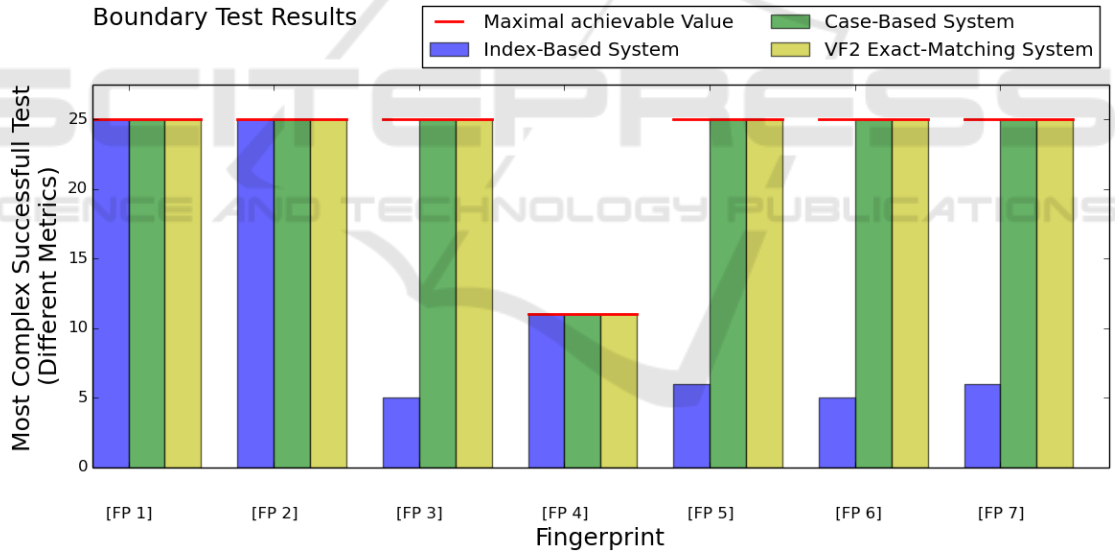


图8：边界测试结果。图中展示了每种搜索技术和每种指纹的边界测试结果。更具体地说，对于每种搜索技术与指纹的组合，该图给出了在具有特定计算能力和内存限制的机器上，系统能够成功处理的最复杂测试场景的复杂度评级。不同指纹的测试场景采用不同的复杂度度量标准。

该表格汇总了总体排名，并展示了针对每个查询的综合结果。排名结果分为两类：第一类为相应检索方法被判定为明显胜出的查询，即所有参与者均认为其为最佳方法；第二类与第一类类似，但显示的是在多少次排名中该方法获得了多数票。

特定查询。在票数相等的情况下，相应的查询被放置在第三列。最后一列给出了由该方法占优的查询所占的百分比。

表1显示，三种检索方法在某些查询场景下均表现良好。在图9中，我们针对每种检索方法，展示了三个选定查询中的两个最佳结果，其中对于这三种检索方法中的每一种——

Table 1: Accumulated results of the result sets for the selected queries. The first category represents queries for which the corresponding retrieval method was considered best by all participants. The second category shows which method won the majority of rankings for the particular query. In case of equality of votes, the corresponding query is placed in the third column. The last column contains the percentage of queries dominated by the particular method.

Retrieval Method	Queries won	Queries won by majority	Co-winner in	Summarized results
VF2	Q3, Q4	Q6, Q7	Q8, Q10	50%
Index-Based	–	Q5	Q10	15%
MetisCBR	Q2, Q9	Q1	Q8	35%

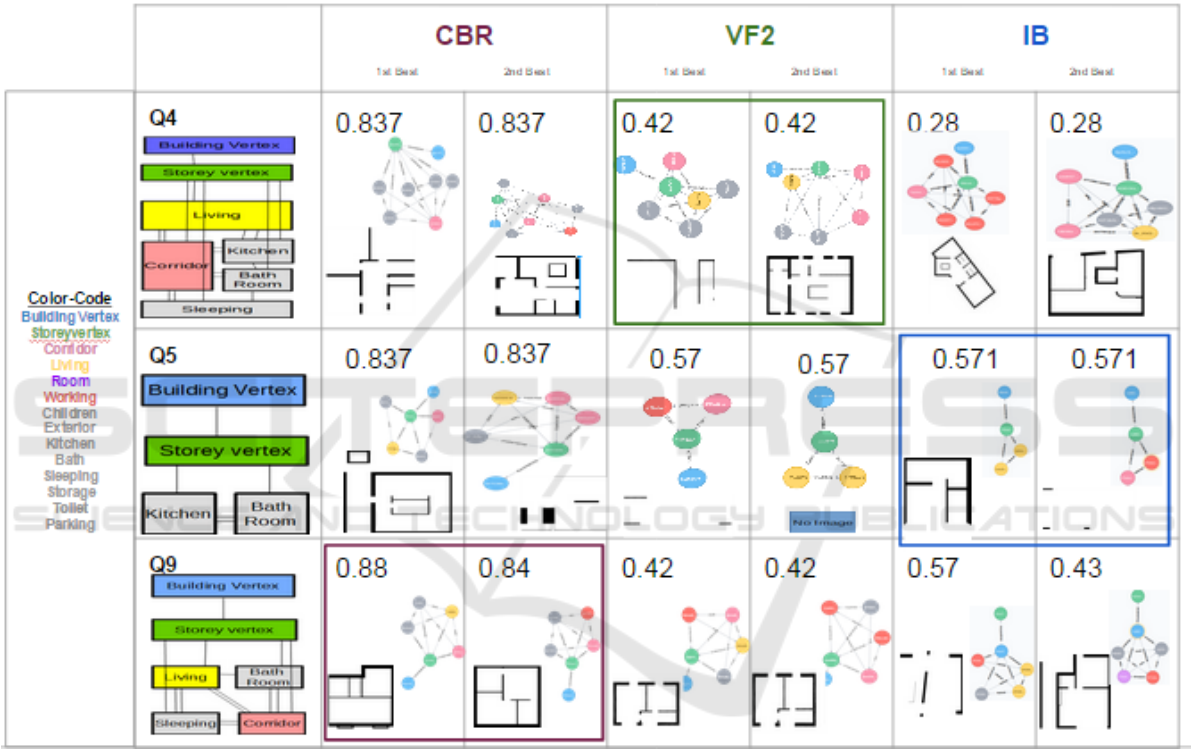


Figure 9: Computed similarity values of the result sets for the selected queries. Color codes represent the room purposes, the first column contains the queries with rooms and assigned purposes. For each retrieval method the first two best single results for the corresponding query are represented. Each single result box consists of a confidence score, the corresponding graph representation with color codes, and the graphical floor plan representation. The result set considered best by the participants is enclosed in a colorized box.

lected queries the best case of each of three retrieval methods is represented, respectively, according to Table 1. For better understanding, how the query and the results looked like and which representations the architects would use while working with our system, we visualized them using both graph-based (the top representation in Table 1) and graphical (the bottom representation in Table 1) representations. This figure does not include the subjective rating of the results, only the confidence scores computed by the corresponding retrieval method are presented. It is notable that the CBR method has a higher score in relation to other two methods. This can be explained with the fact that the application of some fingerprints (e.g., Room Purposes, Room Count or Edge Count) caused the finding of the (almost) exact matches of the room label sets or number of the contained rooms or room connections. This increased their confidence in the accumulated result set. This difference in scoring is observable in other result sets as well.

All things considered, the retrieval methods were able to present sets of reasonable results, taking the

表1：所选查询的结果集累积统计结果。第一类为所有参与者均认为相应检索方法最优的查询；第二类显示在该查询中获得多数排名的方法；若出现票数相同的情况，该查询被列入第三列；最后一列给出由该方法占据优势的查询所占百分比。

Retrieval Method	Queries won	Queries won by majority	Co-winner in	Summarized results
VF2	Q3, Q4	Q6, Q7	Q8, Q10	50%
Index-Based	–	Q5	Q10	15%
MetisCBR	Q2, Q9	Q1	Q8	35%

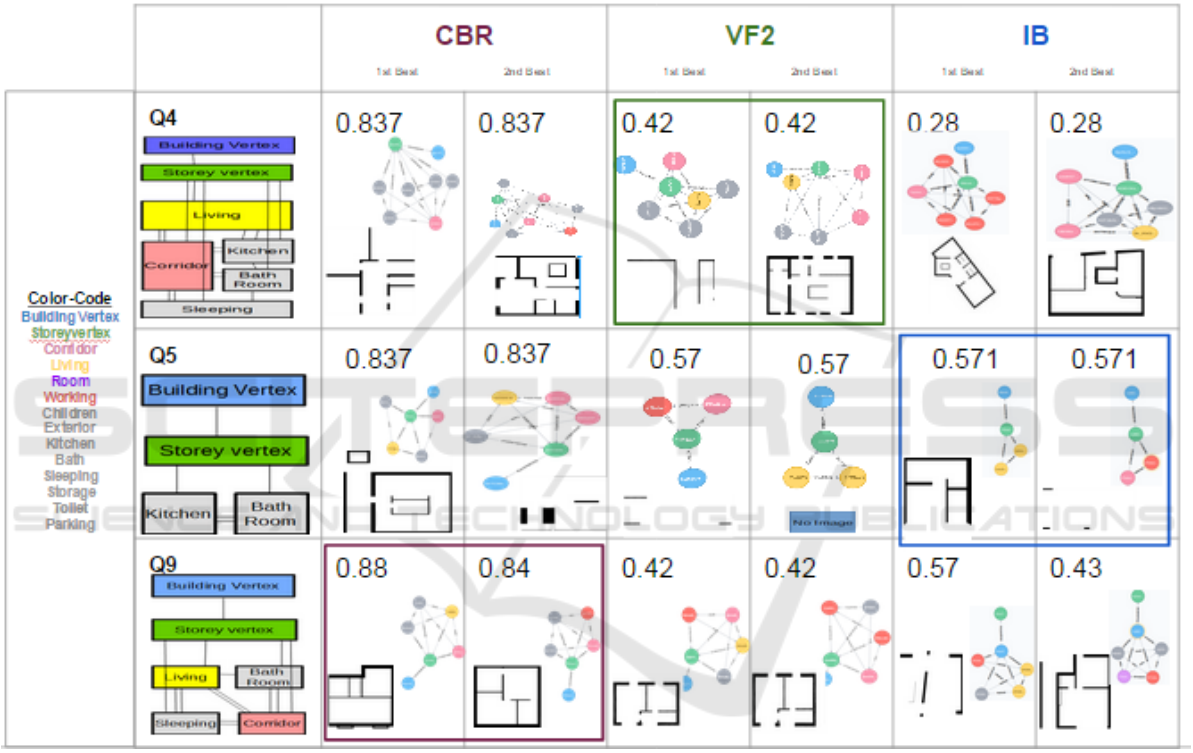


图9：所选查询结果集的计算相似度值。颜色编码表示房间用途，第一列为包含房间及其指定用途的查询。对于每种检索方法，均展示了对应查询的前两个最佳单一检索结果。每个单一检索结果框由置信度分数、带有颜色编码的相应图谱表示，以及平面布局的图形化表示组成。参与者认为最优的结果集被彩色方框圈出。

在所选查询中，三种检索方法的最佳结果分别如表1所示。为便于理解，我们通过基于图的表示（表1上方）和图形化表示（表1下方）直观展示了查询及其结果的外观，以及架构师在使用本系统时可能采用的表示方式。该图未包含对结果的主观评分，仅呈现由相应检索方法计算出的置信度得分。值得注意的是

CBR方法相较于另外两种方法获得了更高的评分。这一现象可归因于：某些特征指纹（如房间用途、房间数量或边数）的引入，使得系统能够识别出与目标房间标签集合、所含房间数量或房间连接关系“几乎”完全匹配的结果，从而提高了累积结果集的置信度。这种评分差异在其他结果集中亦有体现。

综合考虑，检索方法为
a能够给出一组合理的结果，同时考虑

current stage of the respective method development into account. According to the subjective rankings of results and in contrast to the computed confidence score, the VF2 exact matching method was the best one in the overall rating. This fact supports the assumption that exact matching is a better method for determination of isomorphism in graph-based datasets, where a set of tolerable technical limitations is given.

5 CONCLUSION AND FUTURE WORK

In this paper, we introduced Archistant, a complete end-to-end system for search and retrieval of similar floor plans for architects. By using this system, we now have a complete automated pipeline for sketching architectural concepts and searching for similar ideas. We showed the capabilities of the implemented retrieval techniques by means of both a boundary test study (in order to show fundamental performance capabilities and limitations of the methods) and a qualitative study (in order to show that the methods produce reasonable results). In the boundary test study, the VF2-based and MetisCBR replied flawlessly and the index-based method at least answered most queries without errors. In the qualitative analysis, each method showed at least in some situation good performances. The currently implemented fingerprints mainly take graph-based floor-ground properties into account. Geometric properties are not considered. But these information have a big potential for making search results more accurate. A large-scale user study resulting in a big ground-truth database may be conducted in future and may hereby lead to the deployment of machine learning for automatically selecting the best approach for every search scenario in future.

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考虑到各方法开发所处的当前阶段，根据结果的主观评分，并与计算得到的置信度得分进行对比，VF2精确匹配法在总体评价中表现最佳。这一事实支持了如下假设：在存在一组可接受的技术限制的基于图的数据集中，精确匹配是判定同构关系更为优越的方法。

5 结论与未来工作

在本文中，我们提出了Archistant，这是一个面向建筑师的、用于相似平面图搜索与检索的完整端到端系统。借助该系统，我们现已构建了一条从建筑概念草图绘制到相似方案检索的全流程自动化管线。我们通过一项边界测试研究（旨在揭示各方法的基本性能特征及其局限性）和一项定性研究（旨在验证这些方法能够产生合理的结果），展示了所实现检索技术的能力。在边界测试中，基于VF2的方法和MetisCBR均能无误响应，而基于索引的方法至少也能在大多数查询中避免错误。在定性分析中，每种方法在某些场景下均表现出良好的性能。目前实现的指纹特征主要考虑了基于图的楼层—地面属性，而未纳入几何属性；然而，这些信息具有显著提升检索精度的潜力。未来可开展大规模用户研究，以构建大型真值数据库，并在此基础上引入机器学习，从而为每个检索场景自动选择最优方案。

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